LAB 5

SERVOMOTOR OPEN LOOP FREQUENCY RESPONSE AND TRANSFER FUNCTION ESTIMATION

1. LAB OBJECTIVE

The purpose of this lab is to experimentally determine the frequency response of a DC servomotor system. Experimental data will be obtained to create a Bode plot for the open loop frequency response of a DC servomotor and determine the transfer function of the system. Once the transfer function for the system is known, suitable controllers for the system can be developed.

In the last lab we put a sinusoidal drive signal into the motor and had a brief introduction to the encoder output. We were building tools we need for this lab.

2. BACKGROUND

2.1. Introduction to the DC Servomotor System

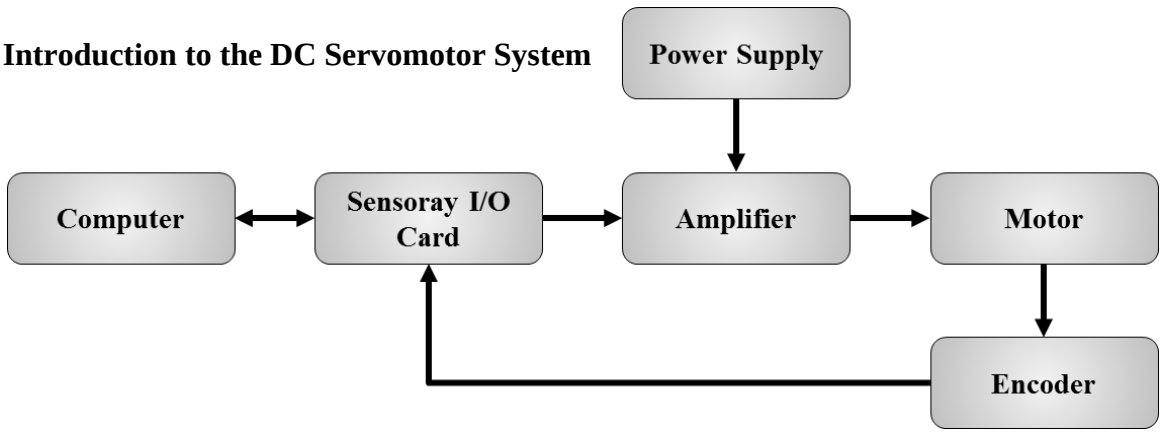

Figure 1. Servomotor drive system schematic

A DC brush-type servomotor is used in the lab. The specifications of the motor and amplifier are as follows.

2.1.1. Bi-directional servomotor

Function:	To supply rotational energy to the system
Manufacturer:	Galil
Model number:	N23-54-1000
Maximum speed:	5900 rpm
T_c - Continuous torque:	0.38 Nm
T_p - Peak torque:	2.95 Nm
Weight:	1.7 Kg
K_t - Torque constant:	0.096 Nm/A
Recommended supply voltage:	60 V
R - Armature resistance:	1.8 Ω

L - Armature inductance:	4.1 mH
T_e - Electrical time constant:	2.27 ms
T_m - Electro mechanical time constant:	8.8 ms

2.1.2. Encoder (comes attached with the motor)

Function:	To detect shaft position
Manufacturer:	Galil
Cycles per revolution:	1000 ppr (pulses per revolution)
Inertia:	2.6×10^{-5} oz-in-sec^2

2.1.3. Servo Amplifier

Function:	To supply power to drive the motor
Manufacturer:	Galil
Model number:	MSA-12-80

Has protection against over-voltage, over-current, over-heating, and short circuits across motor, ground and power leads. Has adjustable gain.

2.1.4. Power supply

Function:	To provide DC supply voltage to the amplifier
Manufacturer:	Galil
Model number:	CPS-12-24
Power rating:	24 VDC @ 12 A

2.1.5. System model

A block diagram of the DC servomotor system transfer functions, which is typical, is shown in Figure 2. The power amplifier shown is a voltage amplifier in which the input and output is a voltage. Servo systems could use current amplifiers instead. The voltage amplifier is modeled as a gain K_a and a lag $(T_a s + 1)$, a low pass filter of sorts. The variable s represents the Laplace operator.

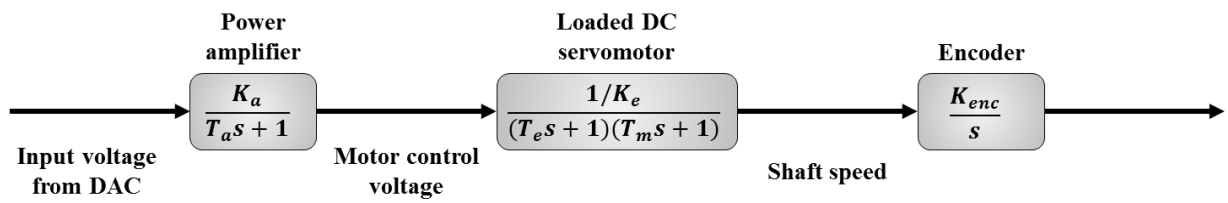

Figure 2. Block diagram of the DC servomotor system transfer functions

The servomotor can be modeled as a second order system. As shown in the Figure 2, the motor gain is the reciprocal of the voltage constant K_e . The electrical time constant T_e is 2.27 ms, according to manufacturer specifications. T_m represents the mechanical time constant of the unloaded motor and is specified to be 8.8 ms. Since the servomotor is connected to an encoder and a coupling load, the actual mechanical time constant T_m is larger than the specified raw value of 8.8 ms.

A counter is required to monitor the output of the encoder. In our case, the Sensoray I/O card has counters that keep track of encoder pulses. The encoder-counter is modeled as an integrator K_{enc}/s, where K_{enc} is the resolution of the encoder and $1/s$ is an integrator represented in the Laplacian domain.

Figure 2 shows an open loop diagram of a servomotor system. The shaft encoder signal can be used as feedback to create a closed loop position control system. For closed loop velocity control, the derivative of the encoder signal would be used as feedback. We will implement closed loop position and velocity control systems in the next lab.

It turns out that the amplifier dynamics $1/(T_a s + 1)$ can usually be neglected. Our system identification data will affirm this. The motor and load can be lumped together as a second order system. The open loop diagram of the servomotor system can be simplified as shown in Figure 3. (Refer to course materials for motor modeling details).

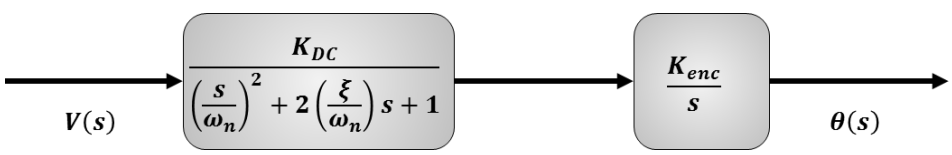

Figure 3. Lumped open loop diagram of a DC servomotor system.

2.2. Second Order Servo System Model:

It turns out that the DC servomotor system used in the lab can be simplified even further as we cover in class. An acceptable second order model for this system is shown in Figure 4.

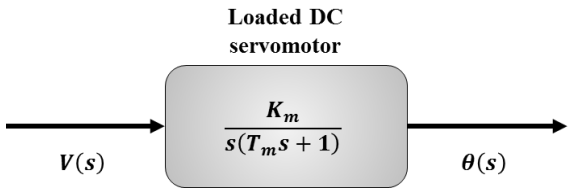

Figure 4. Simplified open loop diagram of the DC servomotor system.

Here, K_m is the net gain of the system and T_m is the electro-mechanical time constant of the system. $\theta(s)$ is the Laplace variable of the angular position and $V(s)$ is the Laplace variable of the input voltage to the system. The Bode plot for this transfer function is shown below in Figure 5.

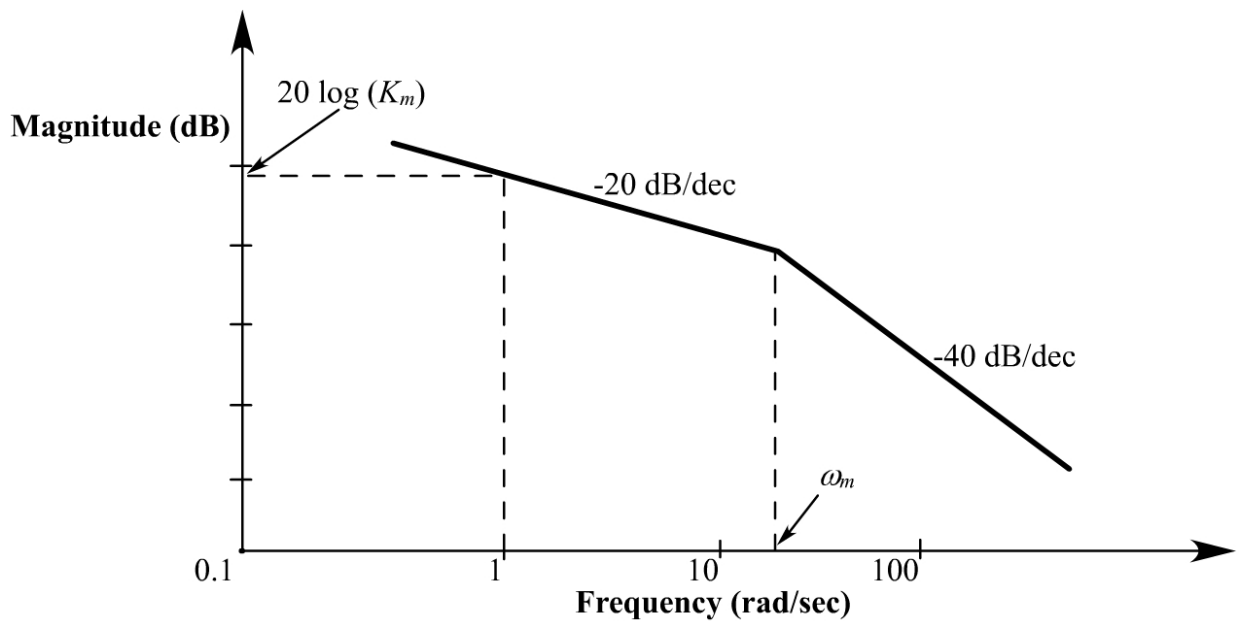

Figure 5. Bode plot for the transfer function $\frac{V(s)}{\theta(s)} = \frac{K_m}{s(T_m s+1)}$

2.3. Experimental determination of the servomotor transfer function

The transfer function magnitude is measured at each of the frequencies shown in Table 1 on page 8. This is done by providing a sinusoidal voltage input to the motor at that frequency and determining the resulting steady state amplitude of the sinusoidal component of the motor's angular position response. The ratio of the output to input amplitude equals the magnitude of the transfer function at that frequency. The data in Table 1 and Table 2 for input amplitudes of 0.5 volts and 1 volts respectively need to be obtained at all the frequencies provided.

You will use the sampling, filtering, and post-processing tools you developed last lab to create the plot from which you will estimate your motor parameters.

2.4. Position sensing

The lab experiments require accurate servomotor shaft angular position sensing. The position sensor that is available in the lab is an incremental encoder-counter arrangement. The incremental

encoder resolution is 1000 counts per revolution. Since the encoder shaft is connected directly to the servomotor shaft, the encoder can detect 360/1000 = 0.36 degree change in motor shaft rotation.

The model 826 board has six identical 32-bit programmable counter channels, numbered 0 to 5. Each counter channel can implement a complete solution for a variety of common applications, including incremental encoder interface, event counter, timer, pulse generator, PWM generator, pulse width measurement, period measurement, and frequency measurement. Each counter channel has six input signals on its header connector: A+, A-, B+, B-, X+, and X-. The header also has power and ground pins that can be used to supply operating power to external devices such as incremental encoders.

You were introduced to the, "SysId" code project last lab and you made necessary changes to suit the process for this lab.

2.5 Drift Removal

We covered this in detail during the last lab. We designed and employed a digital filter. In this lab our lowest command input test frequency frequency is 2.5Hz so consider if you desire to alter your break frequency.

If your high-pass filter attenuates the 2.5Hz due to a break frequency that's too-near this should reveal itself in your data. You will see that perhaps and asymptote through the next couple higher frequency samples has an slope you expect, but it passes (slightly) over the 2.5Hz result. This would indicate possible attenuation of that low-frequency run by your high-pass filter. Comment on this if you see it.

You are also encouraged to post-process filter in Matlab also, given you are logging both raw and high-pass filtered from your C++ code. Consider you have multiple frequency runs for two amplitudes. That's two expected scatter plots.

Now consider processing the raw data from each run through script-time highpass filter in Matlab. You could then plot 2 more data series for comparison. You'd just do something like this on the "raw" data column in your data file. Consider doing this and plotting 4 data series.

```
% high-pass filter...
Fs = 400;  % Hz sampling frequency (CHANGE)
Fc = freq(i)/2;   % Hz cuttoff frequency
N  = 2;    % filter order

t=[0:1/Fs:(M-1)/Fs];

[b,a] = butter(N,Fc/Fs,'high');
yf = filter(b,a,y);

% ignore filter settling samples: 5 seconds worth (CHANGE?)
ignore = Fs*5;

yfi=yf(ignore:M);
C=length(yfi);
tfi=t(ignore:M);
if(plt==1),
    figure('Name','y raw and filtered')
    plot(t,y,'r',t,yf,'b',tfi,yfi,'g')
end
```

2.6 Save and Post-Process Encoder Data

You developed a process for saving output data last week, and for post-processing via Matlab script to estimate root-mean-square and from that output amplitude. With that your could estimate a plant gain at a single frequency.

This lab consider how you will read-in all your files and build a Magnitude plot from which our course material guides us on how to then estimate model parameters from the experiemental data plot.

3. PRELAB ASSIGNMENT

1. Use your SysId project and post processing script from last week. Make desired changes to invoke your post-processor as a function from a wrapper that manages the dozen(s) of data run files you will be generating in lab.

2. Prepare a Matlab script to batch-process your Input files. You will be making two frequency sweeps: one sweep for each of two input amplitudes.

    a. As recommended last week you should design your real-time data logger to impart the amplitude and frequency conditions to each log file (first-row metadata as recommended in lab 4).
    b. With that you can dump all your output data run files into a folder as filename1.txt and so on (any naming convention you desire).
    c. Your post-processing script should then...
        i. Process all files through your RMS-to-amplitude calculation.
        ii. Associate each amplitude estimate with input frequency and amplitude.
        iii. 2-series scatter plot: one series for each voltage level.
            1. $Y=20*\log10(\theta_{out}/V_{in})$ vs semilogx angular frequency
                a. Mind your Hz and radians.
                b. Given 1,000 encoder counts per revolution you can get θ from 'counts'.
    d. With this script be prepared to draw Bode asymptotes on each data series that you will use to estimate plant parameters K_m and T_m.
3. Given our Transfer Function in 's' and thereby 'jω' for steady-state frequency input:

$$H(s) = \frac{\theta(s)}{V(s)} = \frac{K_m}{s(T_m s+1)}$$

$$H(j\omega) = \frac{K_m}{j\omega(j\omega T_m+1)} = \frac{K_m}{j\omega - T_m\omega^2}$$

PRELAB SUBMIT Hand-work derivation of the magnitude expression: $M(\omega) = |H(j\omega)| = \frac{K_m}{\sqrt{T_m^2\omega^4+\omega^2}}$

PRELAB SUBMIT Hand-work reasoning as to why you will be able to read $\mathbf{K_m}$ off the Log-magnitude plot asymptote through your experimental data for ω=1. (Manipulation and approximation of the M(ω) expression above.

PRELAB SUBMIT YOUR DRAFT SCRIPT ABOVE. If you break it into functions just copy-paste into a single document for upload.

4. LAB PROCEDURE

In this lab we will determine the frequency response of the plant by applying a sinusoidal input voltage and recording the amplitude of the motor shaft position as a function of frequency.

First, connect the DACO channel to the input of the servomotor. In your program (tested with the oscilloscope first) set the drive frequency to be equal to 2.5Hz and set the amplitude of the driving signal to be 0.5V amplitude. Execute your program. Using your Matlab code from prelab exercise 3, calculate the peak-to-peak value of shaft oscillations. Write down this value. Repeat the above procedure for the different drive frequencies listed in the tables at the end of the lab handout.

Repeat the experiments for 1V amplitude sine wave input.

5. POSTLAB ASSIGNMENT AND LAB REPORT

1. Create Bode plots (magnitude only, log-log plots) of the open loop frequency response of the plant. Use the data obtained from the experiments with 0.5V and IV sine waves. Plot both plots on one graph for comparison. Discuss the differences and similarities. What do you expect in terms of the similarity of these plots and what did you get? What do you think might be the reason if there is an inconsistency?

2. Based on the frequency response shown by the Bode plots, estimate the open loop transfer function of the plant. Assume the transfer function has the form illustrated in Figure 5. Your plots should be similar to this figure. The gain K_m is the value of the y-ordinate where the experimental curve crosses the $\omega = 1$ rad/s line. The frequency ω_m is the point where the plot transitions from a –20 dB/dec line to a –40 dB/dec line. Then $T_m = 1/\omega_m$. Obtain transfer functions for both 0.5 V inputs and 1.0 V input. Compare the two transfer functions.

POST-LAB SUBMISSION

1. Answers to the two above questions and the plots of Postlab question 1. Show your work for question 2 on these plots.

2. Complete record of measurement data.

Table 1
Experimental results of the open loop plant frequency response for 0.5 V sinusoidal input.

FREQUENCY (Hz)	MOTOR SHAFT DISPLACEMENT (peak to peak counts)	MAGNITUDE 20 log (output/input)*
2.5		
5		
10		
15		
20		
25		
30		
35		
40		
45		
50		
55		
60		
65		
70		
75		

*output = peak-to-peak shaft oscillation in encoder counts, input = peak-to-peak voltage.

Table 2
Experimental results of the open loop plant frequency response for 1.0 V sinusoidal input.

FREQUENCY (Hz)	MOTOR SHAFT DISPLACEMENT (peak to peak counts)	MAGNITUDE 20 log (output/input)*
2.5		
5		
10		
15		
20		
25		
30		
35		
40		
45		
50		
55		
60		
65		
70		
75		

*output = peak-to-peak shaft oscillation in encoder counts, input = peak-to-peak voltage.